TOPIC :ENERGY RESOURCE

RENEWABLE RESOURCES

A renewable resource is a natural resource that can be re-made or re-grown at a scale comparable to its consumption.

GLOBAL ENERGY STATISTICAL YEARBOOK 2016

Global Extreme in 2015

Share of renewable in electricity Production (incl. hydro)
Norway (97.9%)

Share of wind and solar in electricity Production Portugal (24.2%)

What Is Renewable Energy?

- ***<u>Renewable energy</u>*** – "any sustainable energy source that comes from the natural environment."
- Some features of Renewable energy
  - ✓ It exists perpetually and in abundance in the environment
  - ✓ ready to be harnessed, inexhaustible
  - ✓ It is a clean alternative to fossil fuels

"energy that is derived from natural processes that are replenished constantly" – defined by the International Energy Agency

RENEWABLE ENERGY


Sick of oil, want new and clean energy...
RENEWABLE ENERGY
OIL

Renewable energy is growing in importance and popularity

• The desire and necessity to avert irreversible climate damage.

• Increasing oil prices.

• The unreliability of non-renewable resources (e.g., the depletion of oil wells). considering these and other factors, governments worldwide support renewable energy with various incentives. This encourages entrepreneurs to make large-scale investments in renewable energy.

Major Renewable Energy Sources

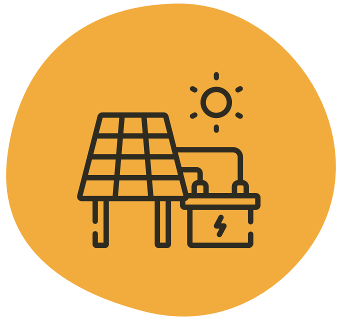
- Hydro energy
- Wind energy
- Solar energy
- Biomass energy
- Tidal energy
- Geothermal energy
- Wave energy
- Bio-fuel
- Biogases

- These sources are normally used to produce **<u>clean</u>** (or **<u>green</u>**) energy. This production does not lead to climate change and does not involve the emission of pollutants.

- A related term is **<u>sustainable</u>** energy: this concept refers to generating energy with an awareness of the future, i.e., in a way that would enable future generations to meet their energy needs too. The concept is related not only to renewable energy but also to **<u>energy efficiency</u>**.

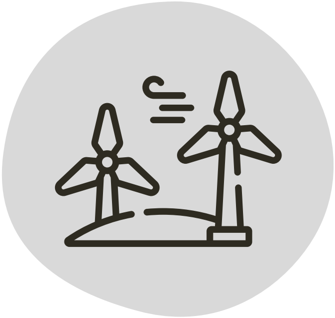
TYPES OF RENEWABLE ENERGY

01



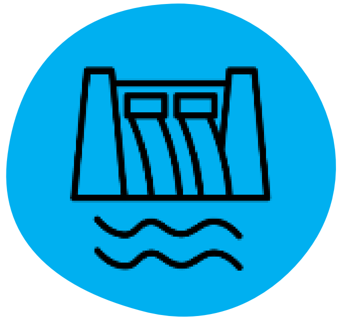
Solar Energy

02



Wind Energy

03

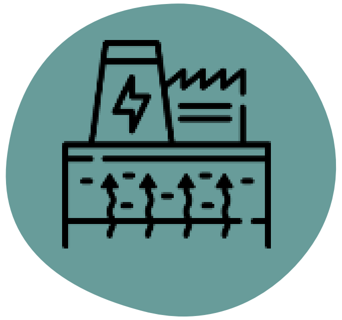


Hydro Energy

04

Biomass Energy

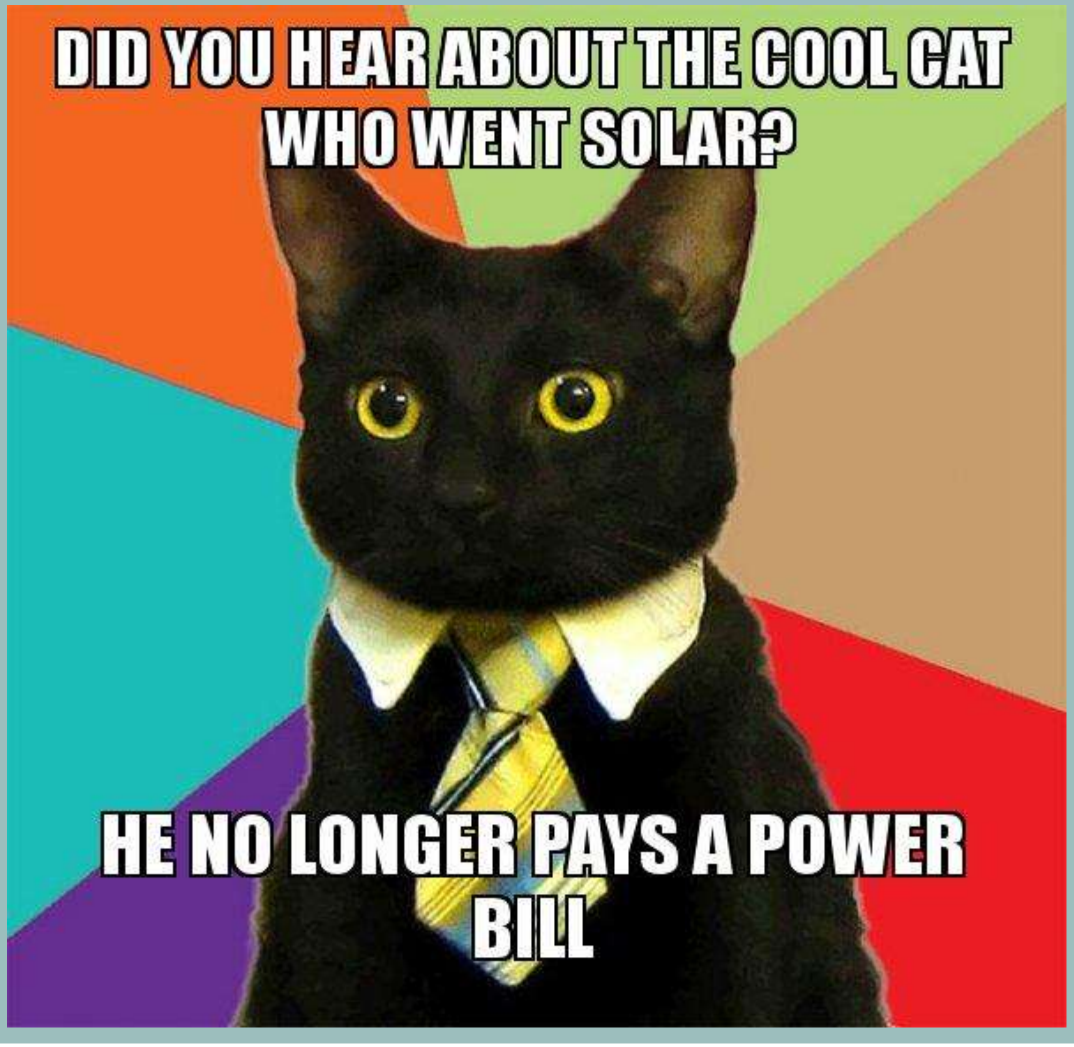
05



Geo Energy

SOLAR ENERGY

Solar power is the conversion of energy from sunlight into electricity, directly using photovoltaic cells, indirectly using concentrated solar power, or a combination. Photovoltaic cells convert light into an electric current using the photovoltaic effect.



There are two main types of solar Energy technologies –

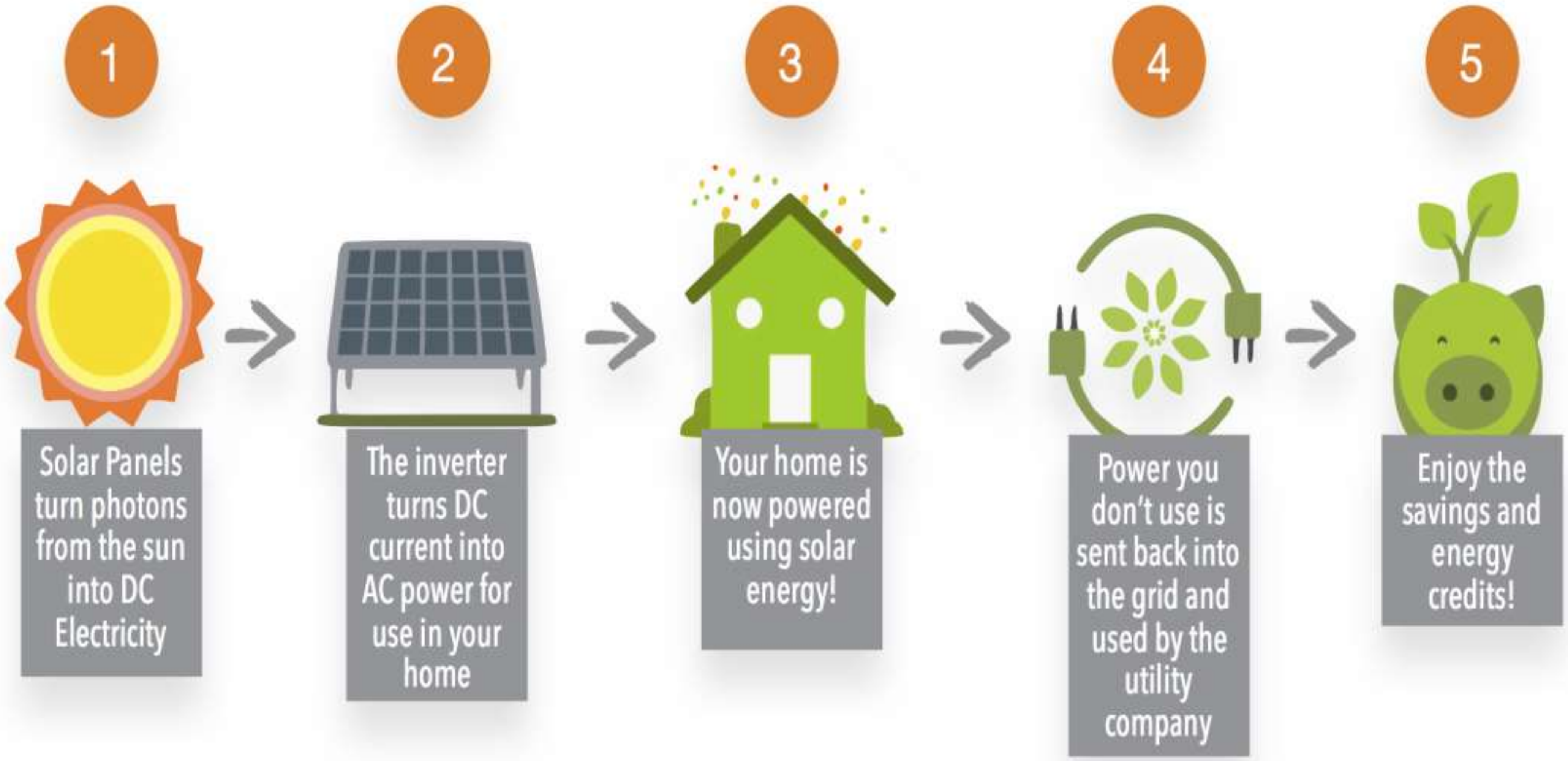
Photovoltaic (PV)

Photovoltaic solar energy is obtained by converting sunlight into electricity using a technology based on the photoelectric effect.

It is a renewable, inexhaustible, and non-polluting energy that can be produced in installations ranging from small generators for self-consumption to large photovoltaic plants.

Concentrating solar-thermal power (CSP)

Concentrating solar-thermal power (CSP) technologies can be used to generate electricity by converting energy from sunlight to power a turbine. Still, the same basic technologies can also be used to deliver heat to a variety of industrial applications, like water desalination, enhanced oil recovery, food processing, chemical production, and mineral processing.

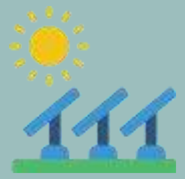

1
2
3
4
5
Solar Panels turn photons from the sun into DC Electricity
The inverter turns DC current into AC power for use in your home
Your home is now powered using solar energy!
Power you don't use is sent back into the grid and used by the utility company
Enjoy the savings and energy credits!

Benefits of Solar Energy:-

Impact on the Environment

Reduce Your Energy Bill

Solar Energy Is Applicable Everywhere



Less Electricity Loss During Long-Distance Transport

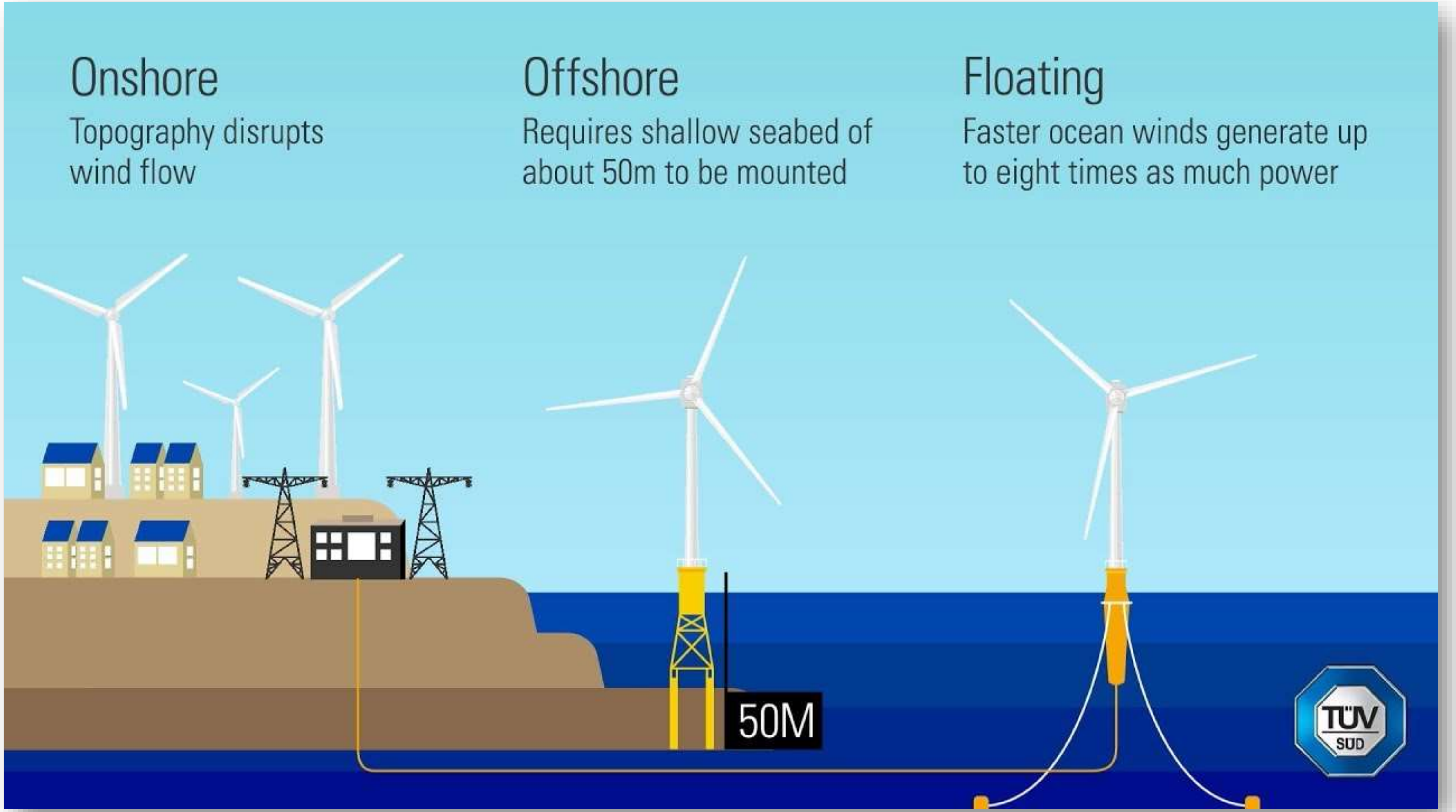
WIND ENERGY

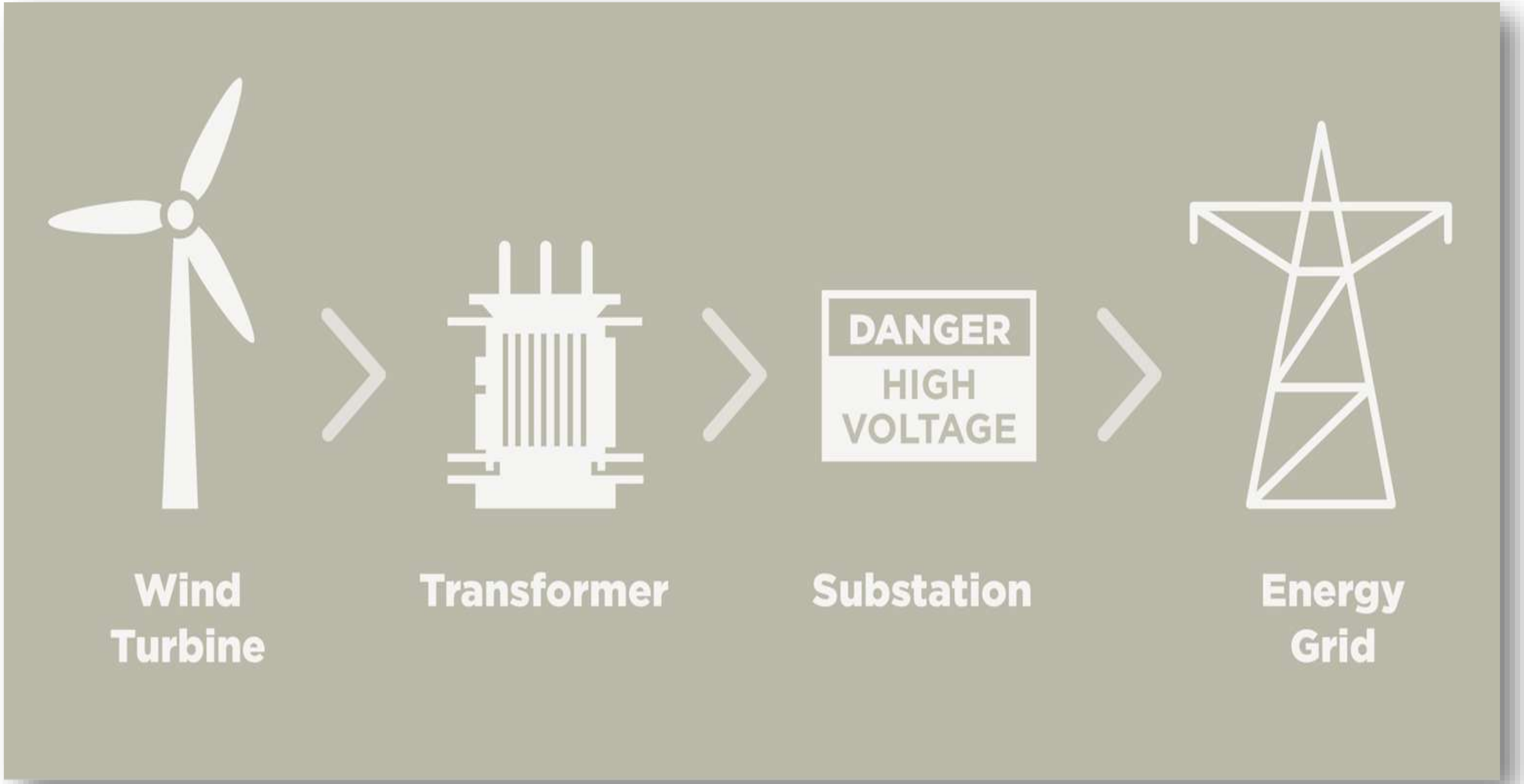
Wind power or wind energy describes the process by which the wind is used to generate mechanical power or electricity.

They are usually grouped in wind farms (sometimes called wind parks).

There are **onshore farms** (which, are often near water); **near-shore farms** (on land or sea within several km of the coast); and **offshore parks** (ten km or more from land).

Tauern wind park, Austria

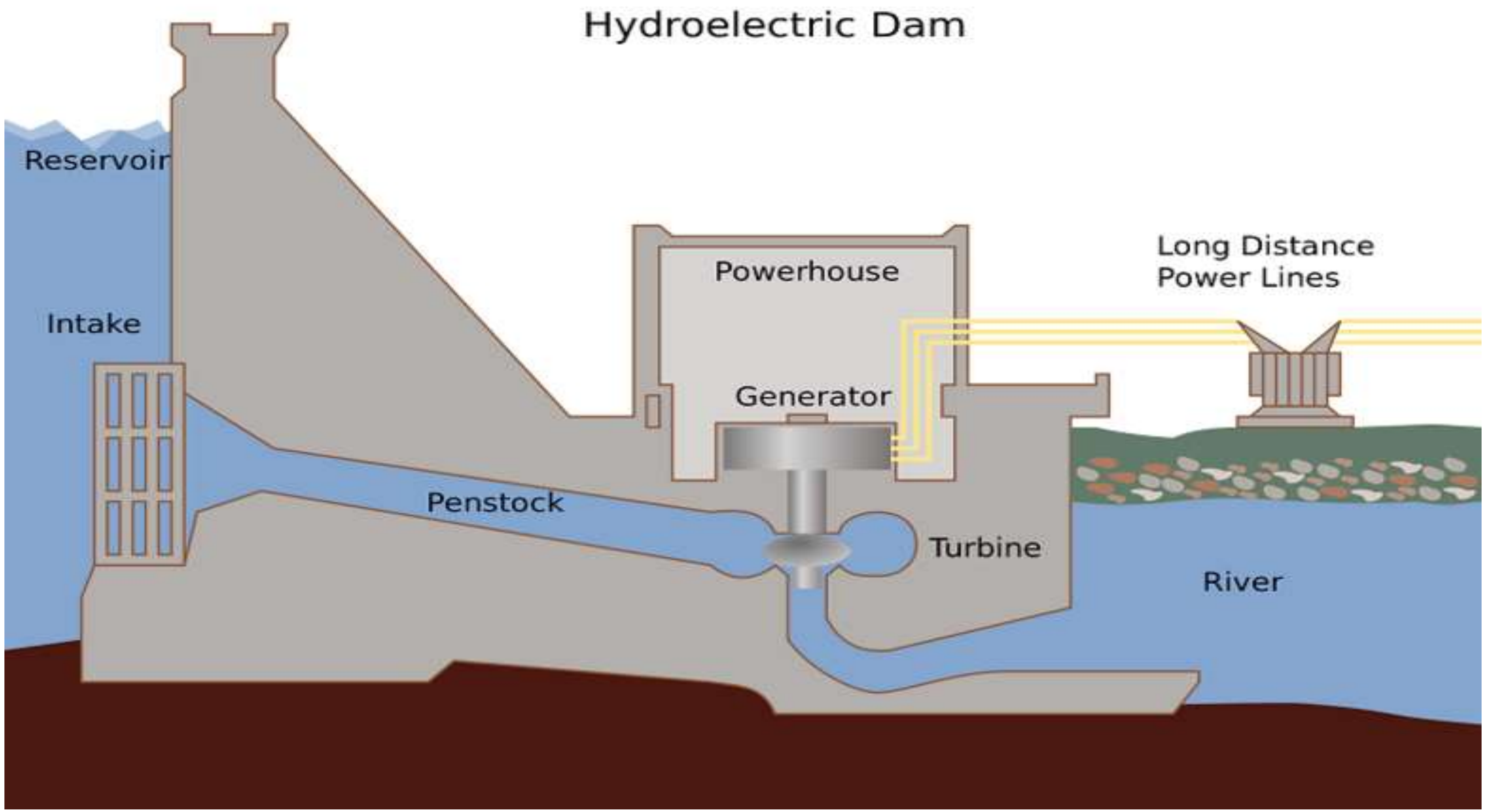

Onshore
Topography disrupts wind flow
Offshore
Requires shallow seabed of about 50m to be mounted
Floating
Faster ocean winds generate up to eight times as much power
50M
TÜV SÜD


DANGER
HIGH
VOLTAGE
Wind
Turbine
Transformer
Substation
Energy
Grid

HYDRO ENERGY

Hydropower, or hydroelectric power, is a renewable energy source that uses a dam or diversion structure to alter the natural flow of a river or other body of water.

HOW DOES HYDRO ENERGY WORK?



Perks of hydro energy-

1. The cheapest energy source

Even though the initial investment required to build a plant is very high and something of a challenge, waterpower is the cheapest source overall in the medium to long term. Once the dams and stations have been built and the turbines installed, they require only minimally expensive maintenance compared to the initial investment.

2. Hydropower stations are agile and responsive

Faced with water availability that can change over time, the power stations are extremely flexible. Production systems require a very small amount of energy to start working.

3. Water brings with it enormous energy

The energy potential of hydropower is enormous. The gigantic masses of water found at high altitudes have a significant amount of gravitational potential energy, and even using just a part of it will provide abundant power.

4. Production can be tailored to demand

The water flow can be easily controlled to suit energy requirements. This is true both in the short term – for example, in the day-night cycle – and in the longer term for seasonal variations and periods of drought.

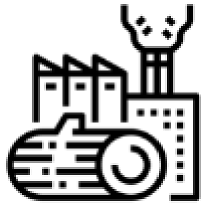
5. Hydropower as a tool for reclamation...

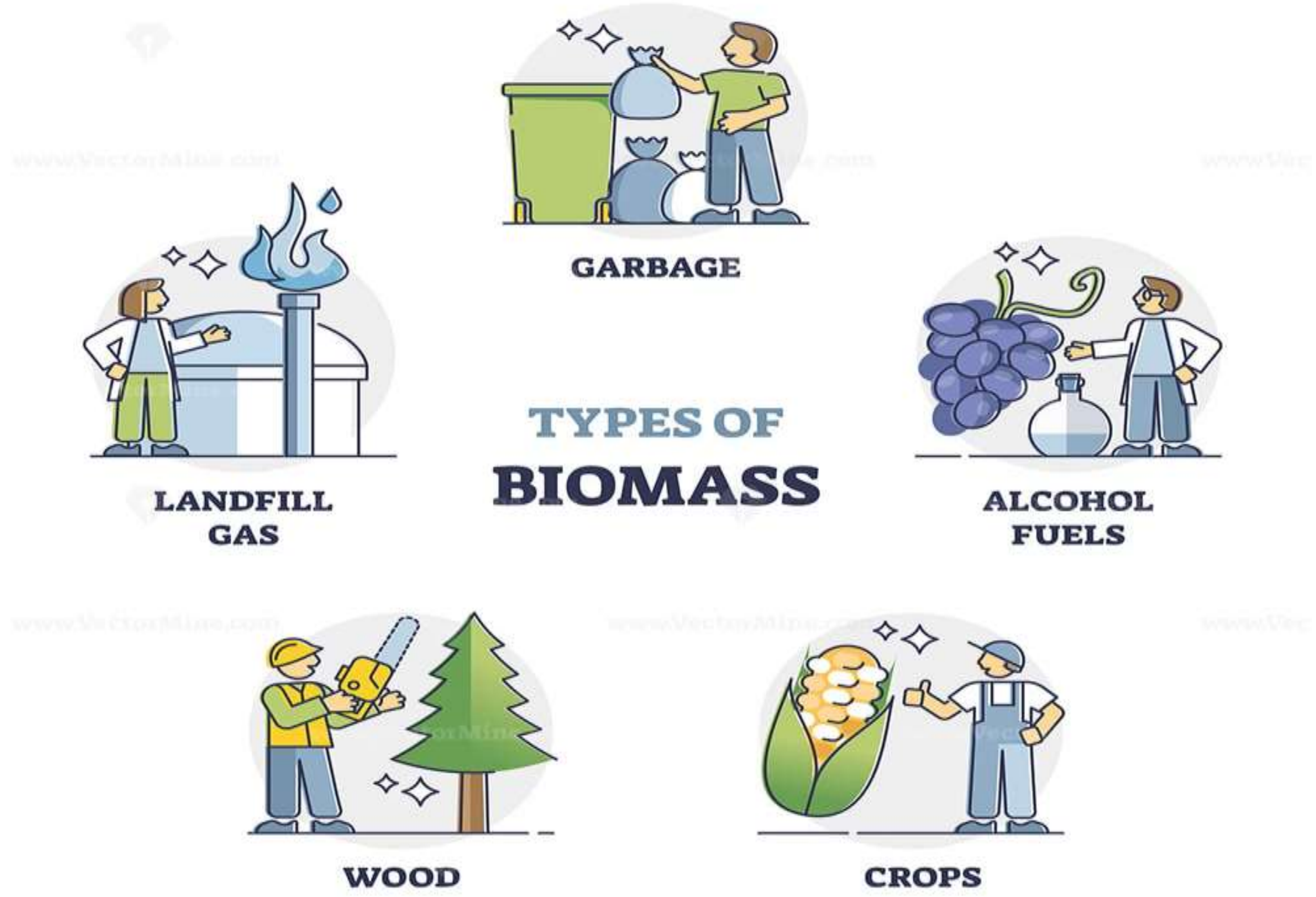
With water courses contained within pre-established reservoirs, hydropower stations also help avoid flooding, to drain marshy land where stagnant water tends to accumulate. The flow and total water volume can be controlled very precisely. Thus, it is possible to ensure that the water flows steadily, even where rainfall tends to be intense and concentrated in short periods.

6 and as an irrigation system

At the opposite end of the spectrum to the previous benefit, hydropower stations can also prove invaluable where water is very scarce. Reservoirs can be used as water reserves for periods of drought. Usually, the water coming out of the station is perfectly clean and usable. The opening of dams can also be planned and calibrated around crop cycles and ecosystems, and the water can be allowed to flow to allow fish to get through at appropriate times.

BIOMASS ENERGY

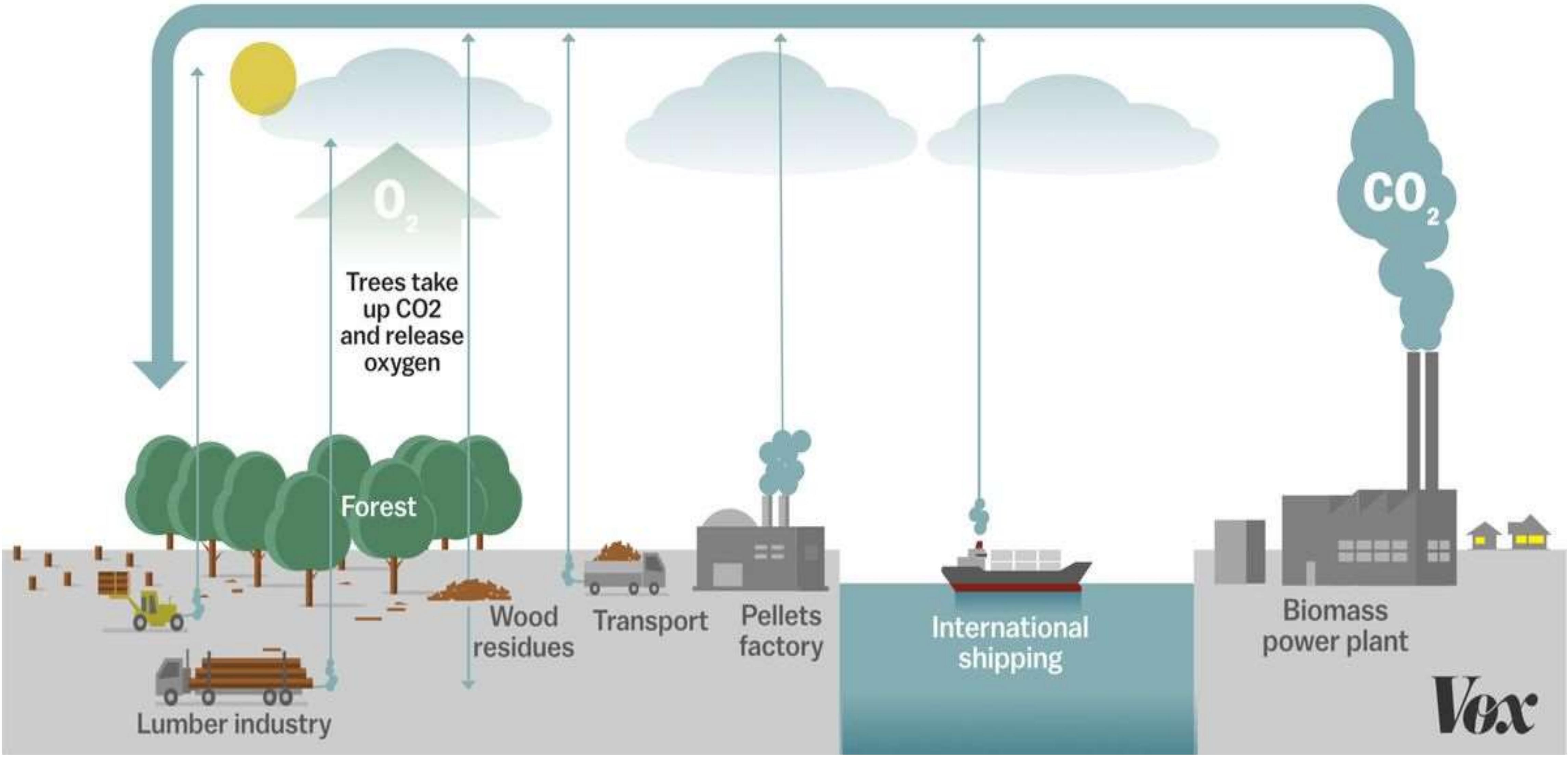


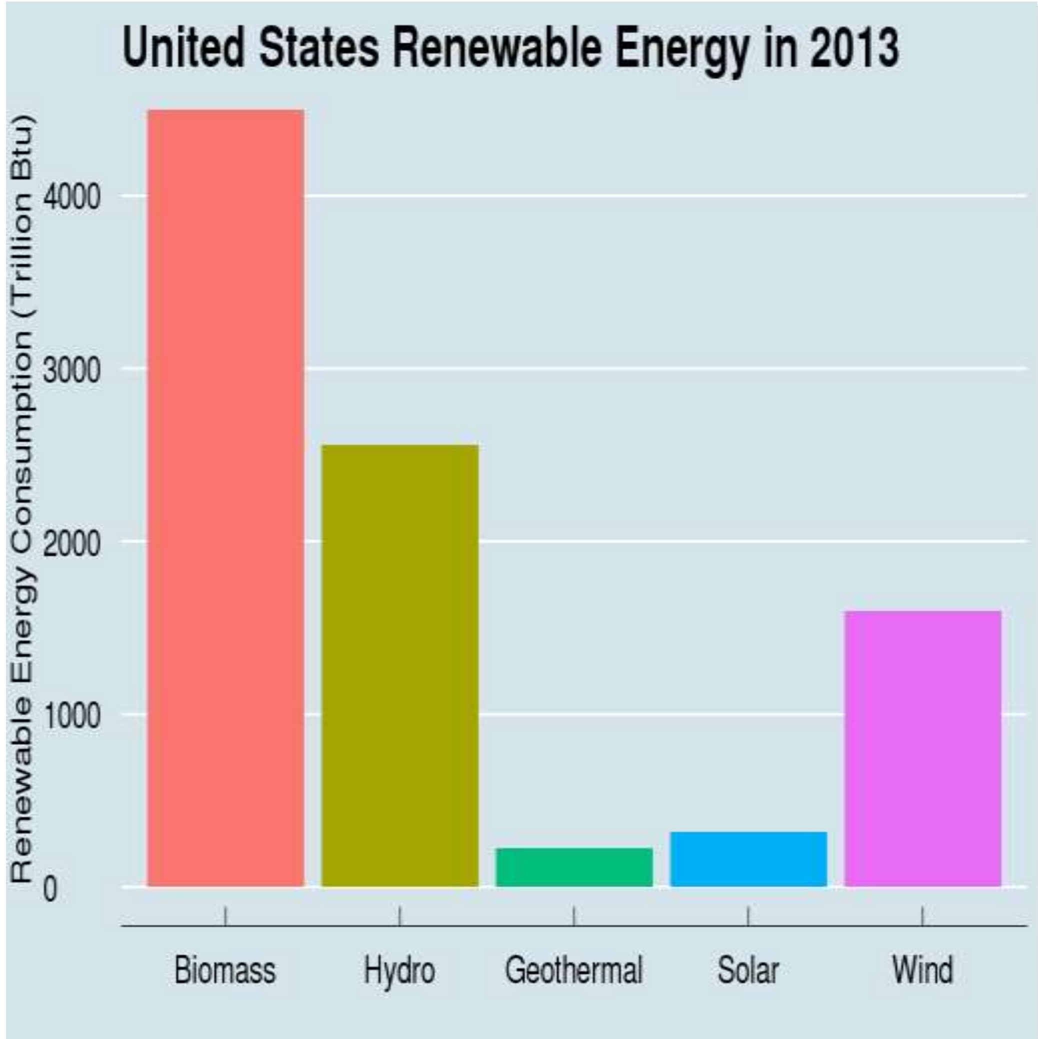


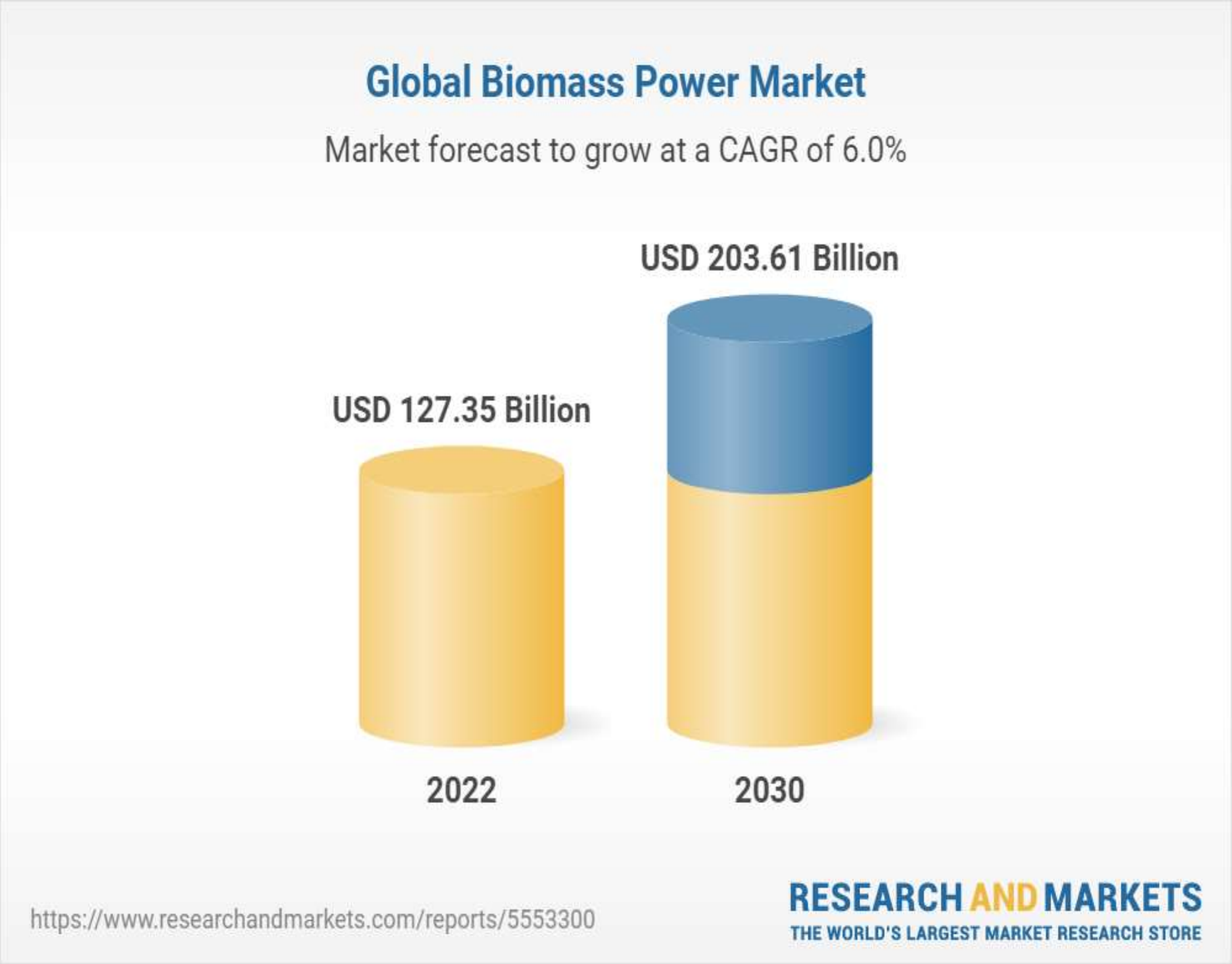
Biomass is a plant-based material used as fuel to produce heat or electricity. Examples are wood and wood residues, energy crops, agricultural residues, and waste from industry, farms, and households. Since biomass can be used as a fuel directly, some people use the words biomass and biofuel interchangeably.

Biomass is probably our oldest source of energy after the sun.

The cycle of biomass energy

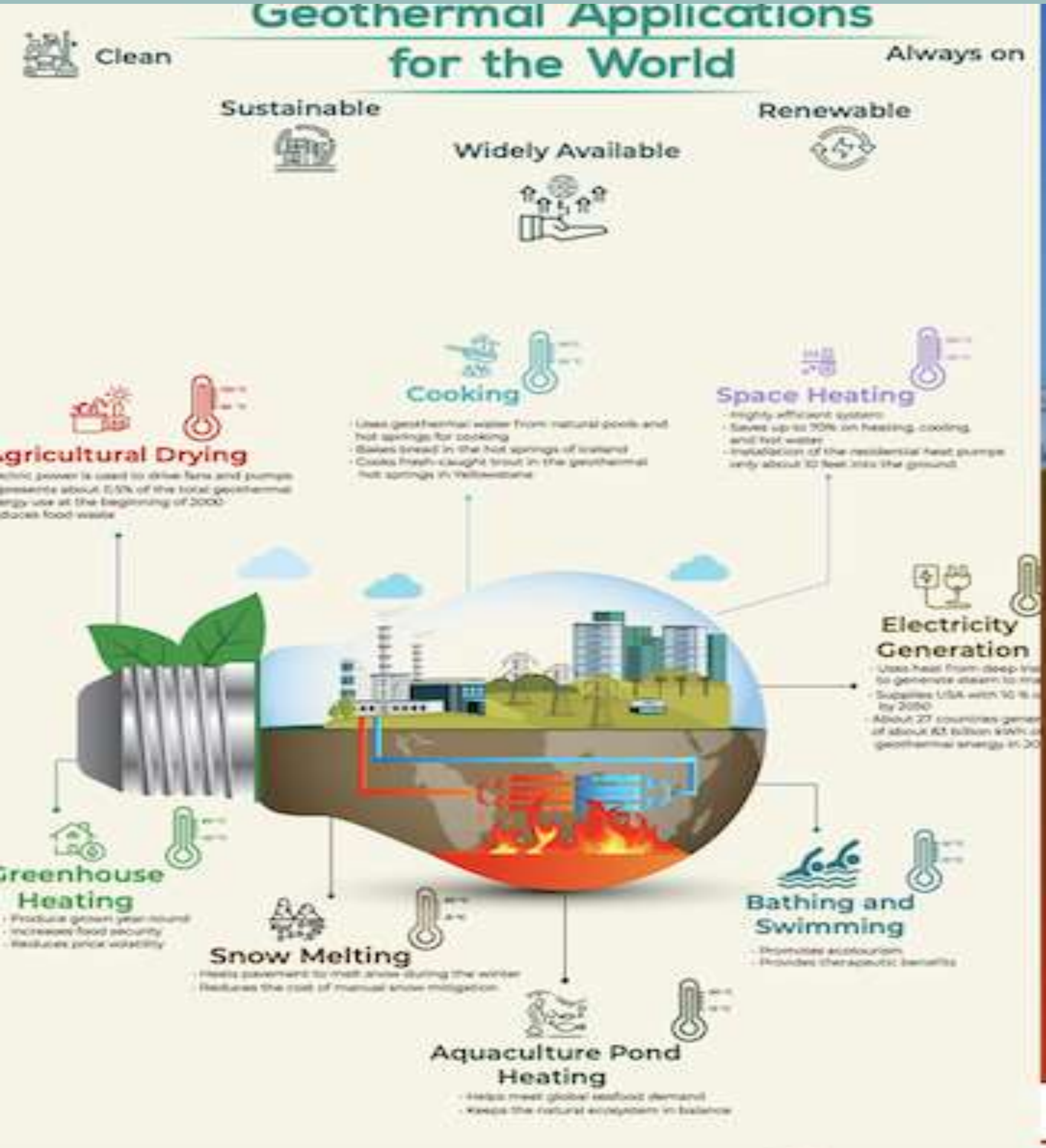


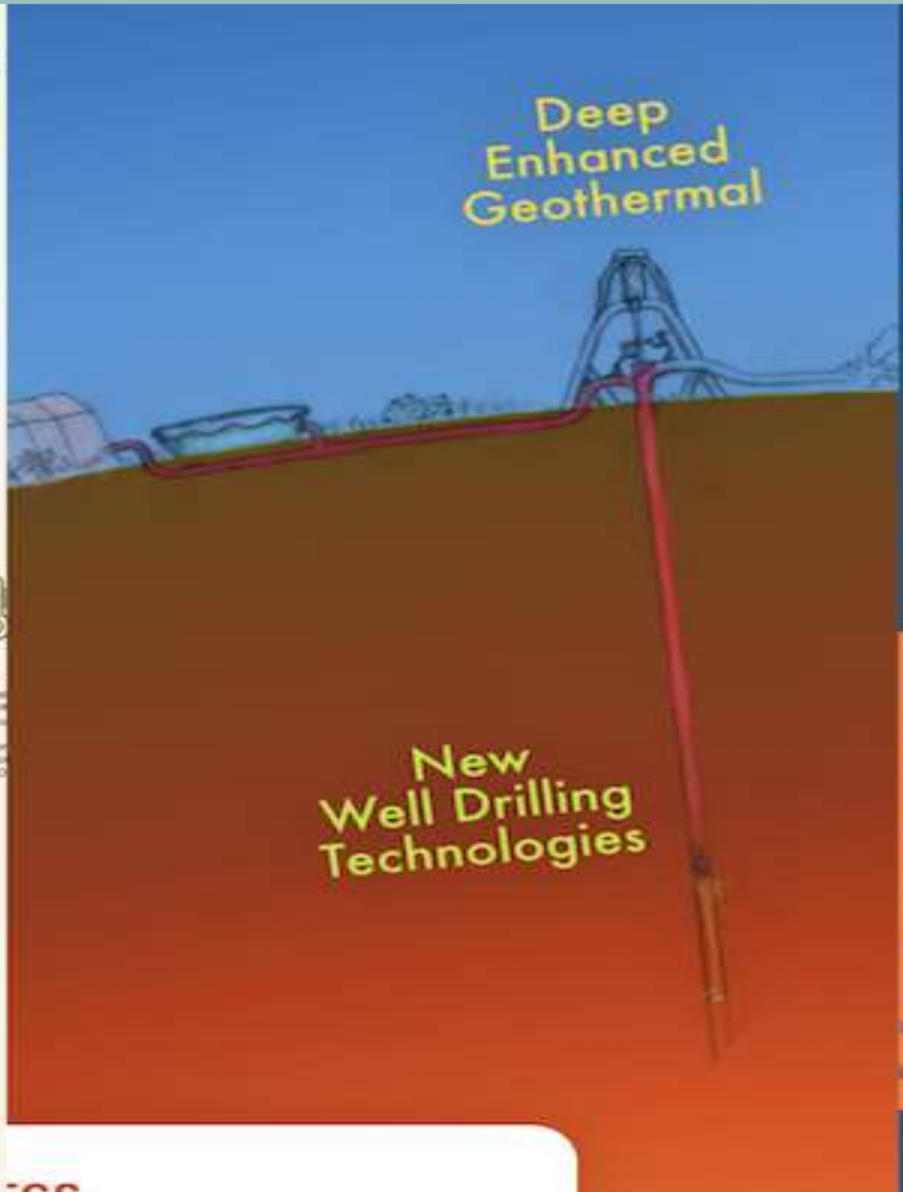

United States Renewable Energy in 2013
Renewable Energy Consumption (Trillion Btu)
4000
3000
2000
1000
0
Biomass
Hydro
Geothermal
Solar
Wind

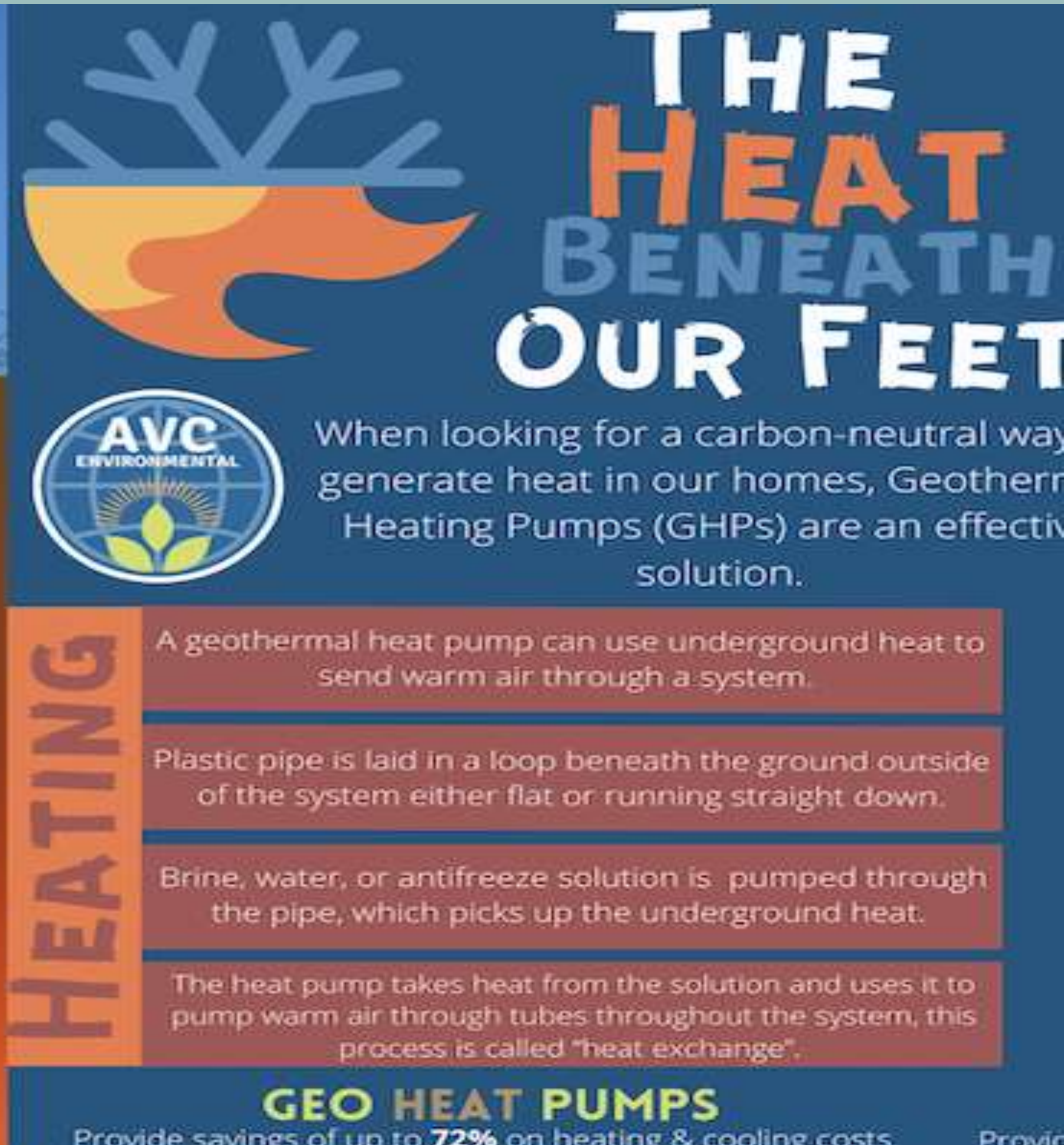

Global Biomass Power Market
Market forecast to grow at a CAGR of 6.0%
USD 203.61 Billion
USD 127.35 Billion
2022
2030
https://www.researchandmarkets.com/reports/5553300
RESEARCH AND MARKETS
THE WORLD'S LARGEST MARKET RESEARCH STORE

GEOTHERMAL ENERGY

Geothermal energy is heat that is generated within the Earth. (Geo means “earth,” and thermal means “heat” in Greek.) It is a renewable resource that can be harvested for human use. About 2,900 kilometers (1,800 miles) below the Earth's crust, or surface, is the hottest part of our planet: the core.







Most small island economies rely on oil-fired power plants to provide a steady electricity supply, but **Iceland** has virtually 100% renewable electricity from its abundant hydropower and geothermal resources.

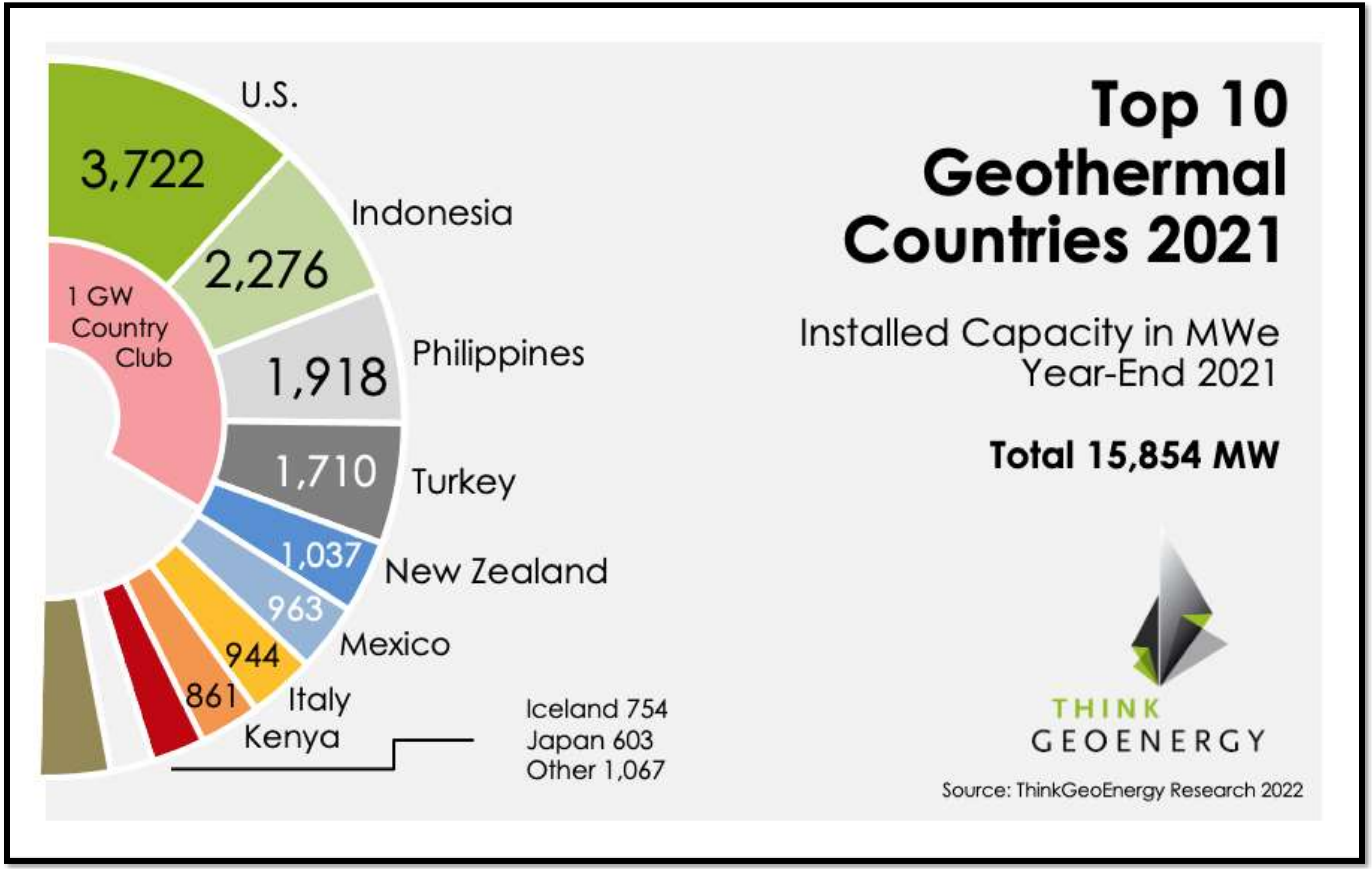
Perks of Geo Energy-

1. Baseload energy - it's always on: Geothermal power plants produce electricity consistently, running 24 hours a day, 7 days a week. The power output of a geothermal power plant is highly predictable and stable, thus facilitating energy planning with remarkable accuracy.

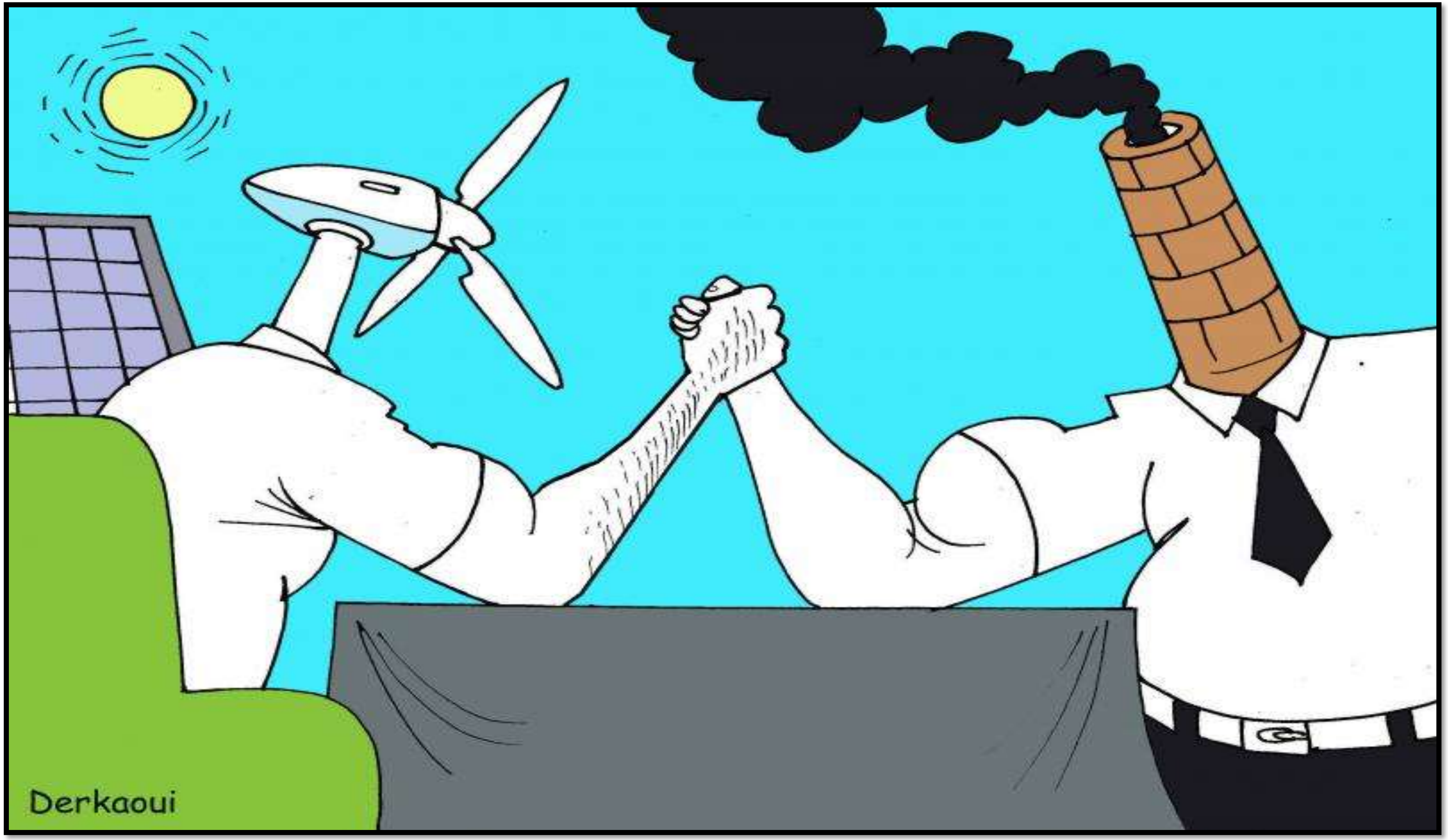
2. Geothermal energy can heat, cool, and generate electricity: Geothermal energy can be used in different ways depending on the resource and technology chosen—heating and cooling buildings through geothermal heat pumps, generating electricity through geothermal power plants, and heating structures through direct-use applications.

3. Over 100 years of geothermal sourced energy: The first ever geothermal plant was set up in Larderello, Italy 1904. Steam from that geothermal source was used to turn a small turbine that powered five light bulbs. Today, the U.S. leads the world in geothermal power generation, providing more than 3.7 GW to the national grid.

4. Weather Independence: A significant advantage is that geothermal energy is not dependent on weather conditions.


Top 10 Geothermal Countries 2021
Installed Capacity in MWe
Year-End 2021
Total 15,854 MW
U.S.
3,722
Indonesia
2,276
Philippines
1,918
Turkey
1,710
New Zealand
1,037
Mexico
963
Italy
944
Kenya
861
1 GW Country Club
Iceland 754
Japan 603
Other 1,067
THINK GEOENERGY
Source: ThinkGeoEnergy Research 2022

WHO WILL WIN THIS BATTLE ?



BENEFITS OF RENEWABLE ENERGY

- ❑ 46M pounds of annual reductions in GHG emissions.

- ❑ 10M gallons in annual water annually savings

- ❑ 225 acres of land preserved by using roofs and parking lots

BENEFITS OF RENEWABLE ENERGY

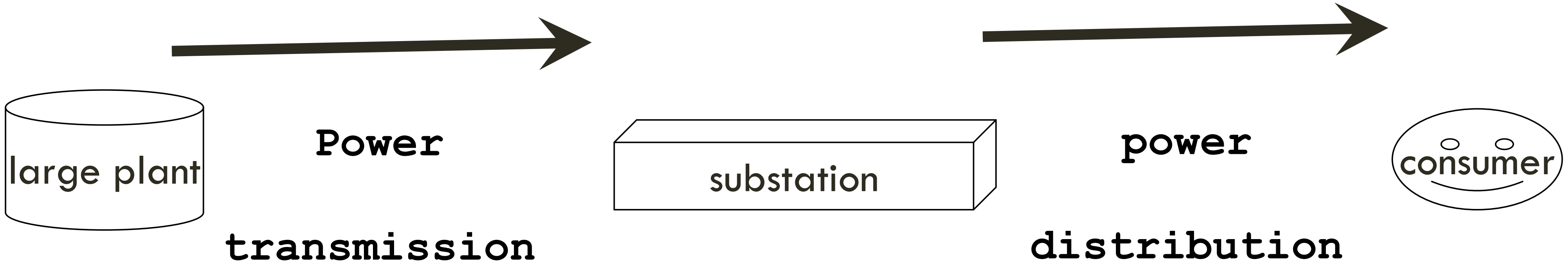
ADVANTAGES & DISADVANTAGES OF RENEWABLE ENERGY

- Easily regenerated
- Supports environment
- Easily available
- Low maintenance cost

- Weather dependency
- High installation cost
- Fluctuation problem(Solar)
- Geographic limitations

Distributed generation

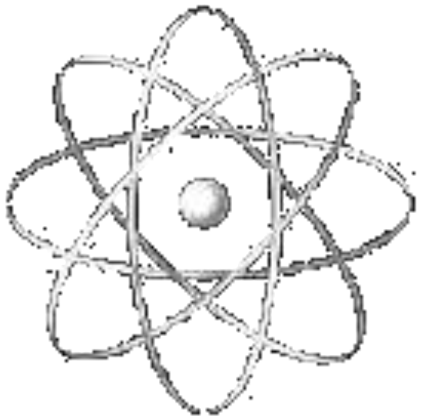
Traditional energy generation is mostly done in centralized facilities, from where energy must travel a long way to reach the end consumer:



By contrast, renewables are often associated with **distributed generation** (also called **dispersed generation** or **decentralized energy**). This is producing energy in many small facilities and transporting it over short distances. Roof solar panels and wind turbines are examples of distributed energy resources (DERs).

For renewable energy to become massively used, energy systems must be adjusted to reflect the shift from centralized to dispersed generation.

What energy qualifies as renewable



Some **scientists and politicians argue** that **nuclear energy is renewable** since the resources from which it is derived (such as uranium) would not be exhausted in millions of years.

These claims, **however, have not been proven;** furthermore, nuclear energy has an extremely dangerous byproduct: **nuclear waste.** For this reason, governments don't recognize nuclear energy as renewable, and it is not eligible for state subsidies.

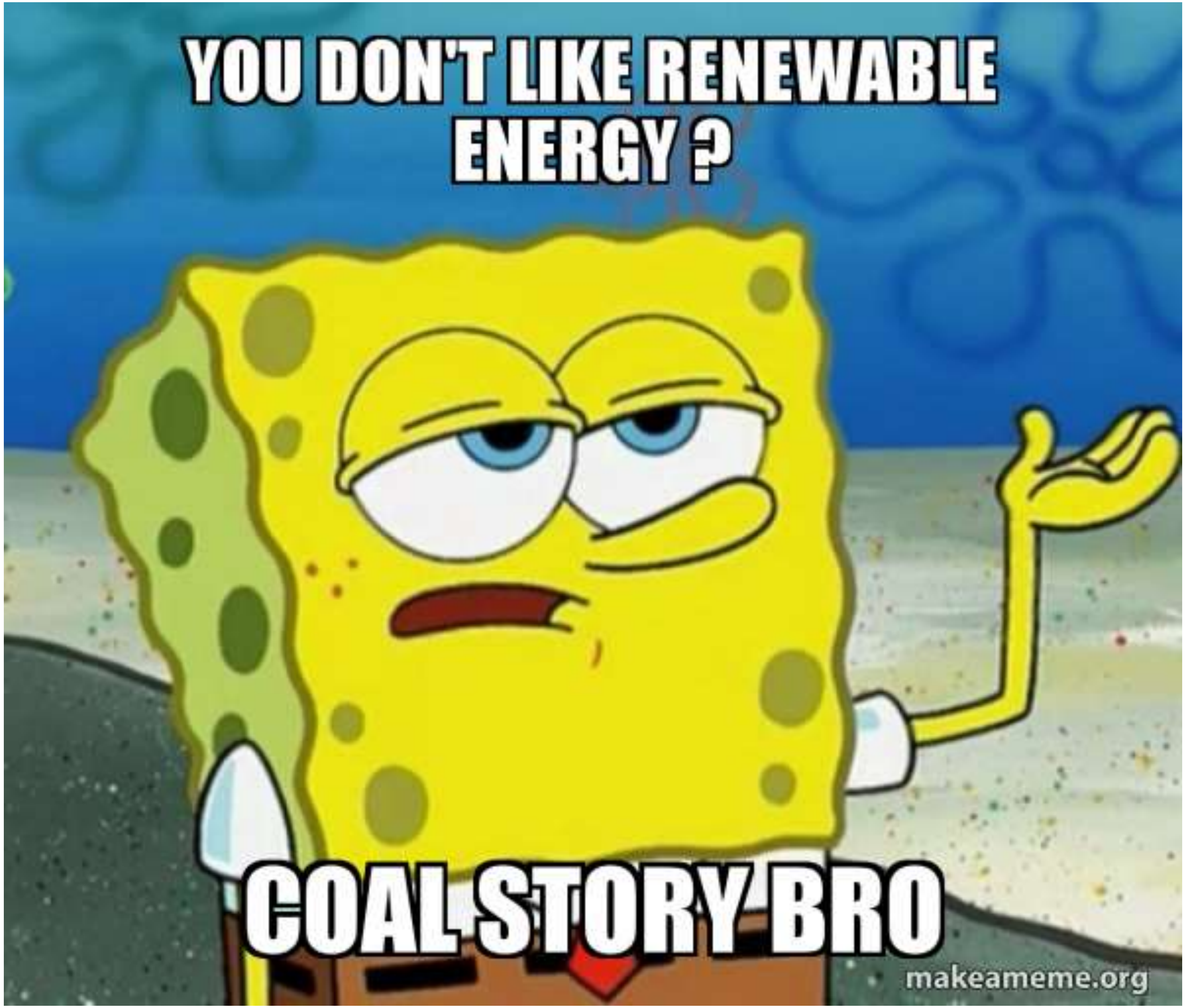
Fossil fuels could be regarded as biomass since they have biological origin; **however, they are neither sustainable nor green because:**

- this is organic material that has undergone millennium-long geological transformation;
- thus, the regeneration rates of fossil fuels are extremely slower than the rate at which they are consumed;
- fossil fuels emit CO_2 when burnt.

Feed-in tariffs

Since renewables are still innovative and in active development, they are often not competitive with traditional energy sources. Therefore, green-minded governments provide various incentives that encourage investments in the sector and promote its faster development. Among the most common is the feed-in tariff. This is an obligation imposed on utilities by the government **to buy a certain amount of renewable electricity at prices higher than the market's rates.** The higher expenditure for the utility is passed on to its customers. The increase in prices that customers must bear is usually small, but these small contributions are a powerful and effective way to support green energy.

Feed-in tariffs were introduced as early as 1978 in the USA. Now, they are implemented in around 50 countries around the world. In Germany, for example, feed-in tariffs are regulated by the Renewable Energy Law (Erneuerbare-Energien-Gesetz). The programme adds around EUR 1 to each monthly residential electricity bill, which translates into billions of euros of subsidies for the energy sector each year. The country aimed to generate 12.5% of its electricity from renewable sources by 2010. The percentage should rise to 20 by 2020.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- https://www.nationalgeographic.org/topics/resource-library-energy-resources/?q=&page=1&per_page=25
- https://www.eia.gov/energyexplained/what-is-energy/sources-of-energy.php
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- https://www.youtube.com/watch?v=PROhNDen3nk